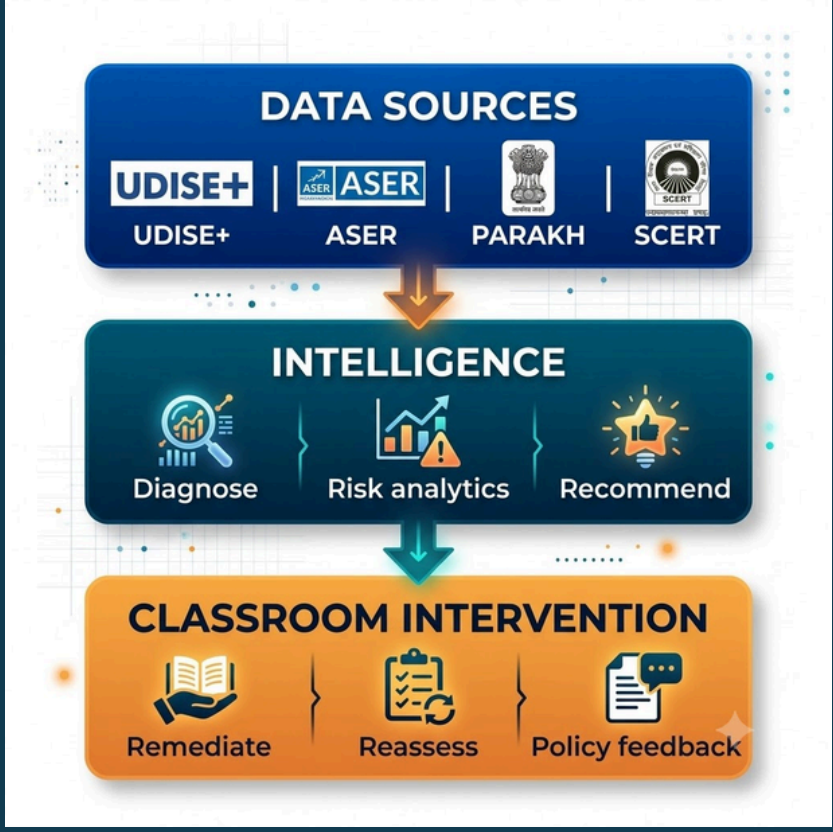
IIT BOMBAY TECHFEST | ROUND 1 PROPOSAL

NIPUNSetu

Bridging India's Foundational Learning Data-to-Action Gap

An Interoperable Decision-Support Platform for Foundational Literacy and Numeracy



India doesn't lack education data. It lacks the conversion of existing data into classroom-level action.

Team Name: AGILE **Team ID: TM-1BEF0BF833D**
Institution: JIIT, Noida
Team Members: Aditya Pandey, Akriti Srivastava

From educational data collection to measurable classroom action.

India's Enrolment Success vs Foundational Learning Gap

The problem

- Access is largely solved: children are in school. Learning, especially reading and basic numeracy, is the gap.
- Data from UDISE+, ASER and PARAKH exists, but it does not reach the teacher as timely, competency-level action.
- There is no closed loop from diagnosis to remediation to reassessment.

A

27.1%

of Std III children can read a Std II-level text (ASER 2024, all schools)

B

23.4%

of Std III children in government schools can do the same (ASER 2024)

C

PARAKH

PARAKH 2024: national Grade 3 assessment evidence

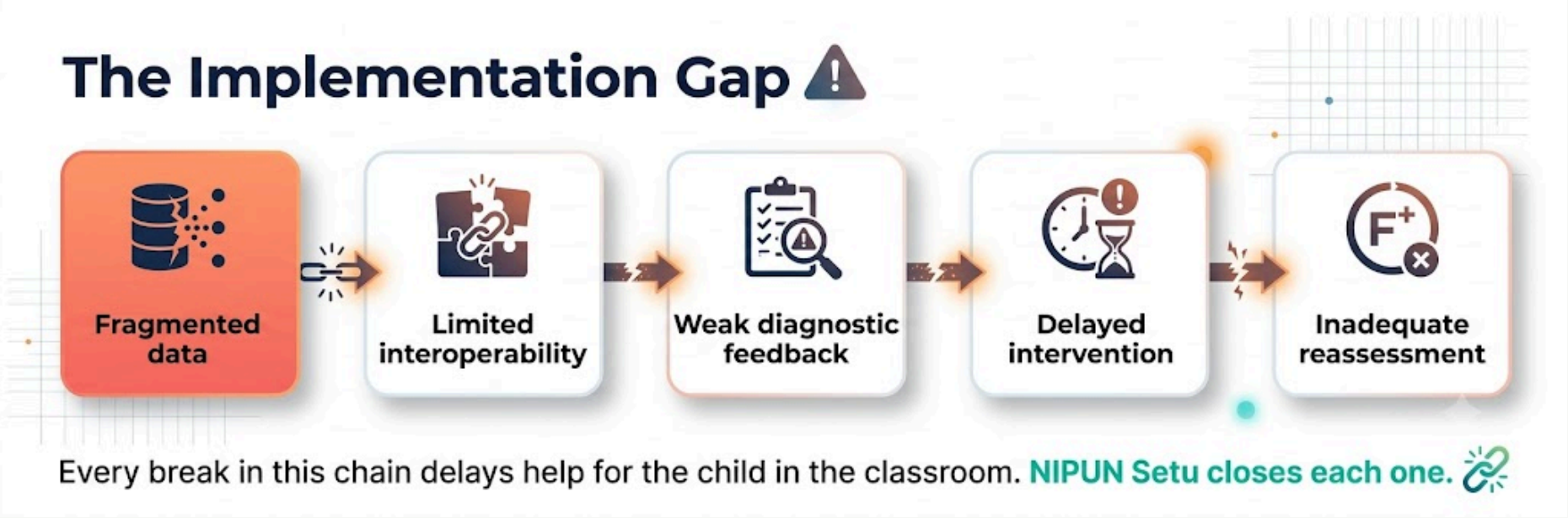
Underserved beneficiaries

Government primary school children, especially Grades 1-3, and the teachers and administrators responsible for them.

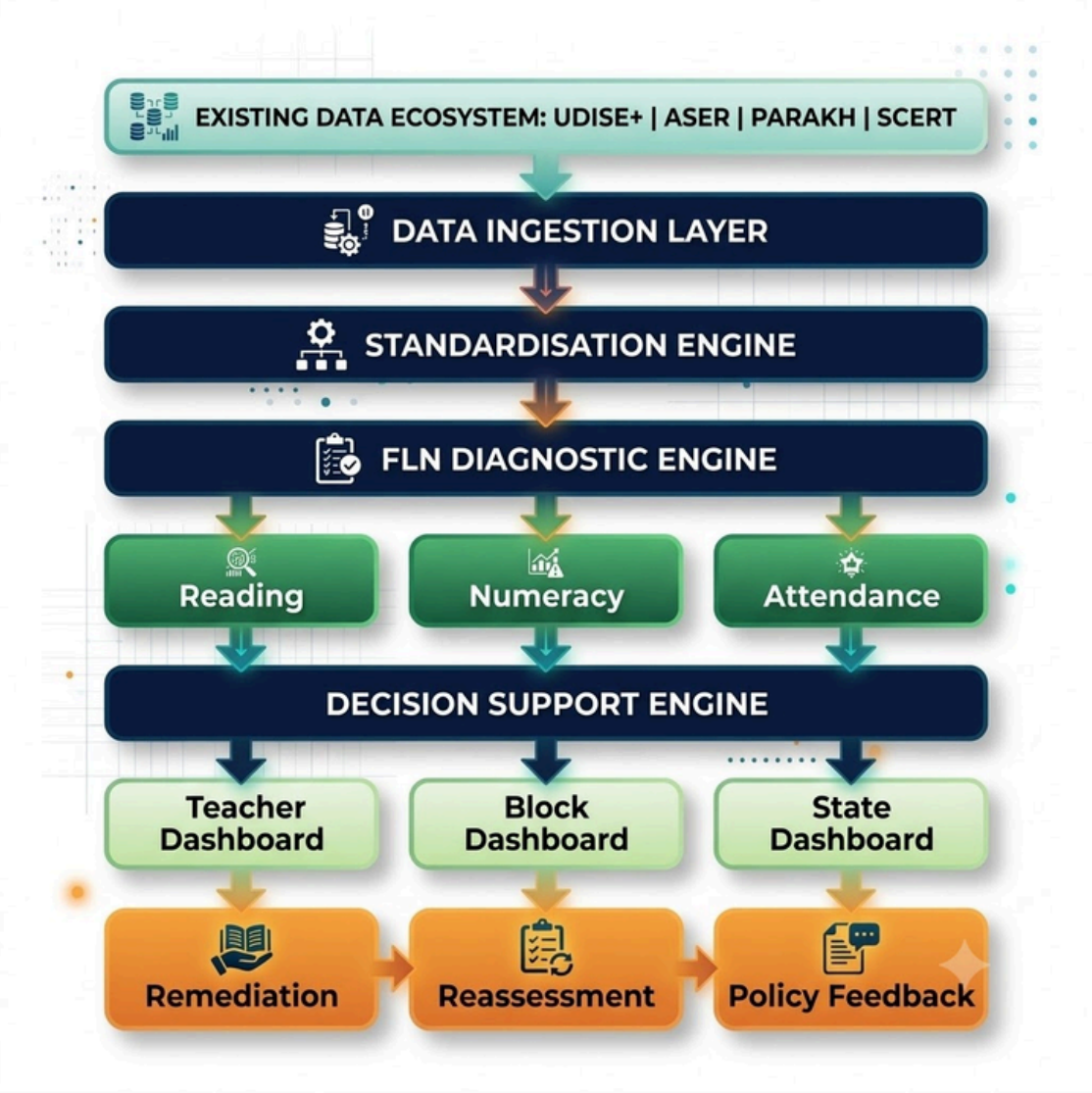
Note: ASER figures are rural household survey estimates, not UDISE+ or school administrative statistics.
Sources: ASER 2024; PARAKH 2024.

India Has the Data. The Missing Link Is Action

Existing initiative	Contribution	Remaining gap
UDISE+	School information	**Limited classroom-level diagnostic action**
ASER	Foundational learning evidence	**Survey-level insights**
PARAKH	Competency assessment	**Translation into continuous remediation**
NIPUN Bharat	FLN policy mandate	**Implementation monitoring**
DIKSHA	Teaching resources	**Contextual resource discovery**



Introducing NIPUNSetu



1 **Data interoperability**

One standard schema across UDISE+, ASER, PARAKH and SCERT data.

2 **Competency-level diagnostics**

Reading, numeracy and attendance signals for every learner.

3 **Explainable learning-risk analytics**

Teachers see why a child is flagged, not just a score.

4 **Longitudinal intervention tracking**

Remediation and reassessment tracked over time.

How We Make It Work

GitHub Link

Architecture & Feasibility Overview:-

A. Technology Stack

Frontend	HTML, CSS, JavaScript
Backend	Python, FastAPI
Database	SQLite prototype; PostgreSQL planned
ML	Scikit-learn
Data	Pandas, CSV, JSON
Authentication	JWT

B. AI/ML Pipeline

1. Assessment History
2. Feature Engineering
3. Learning Risk Analysis
4. ML Prediction
5. Teacher Recommendation
6. Reassessment

⚠ ***Experimental:*** *requires validation on authorised longitudinal data.*

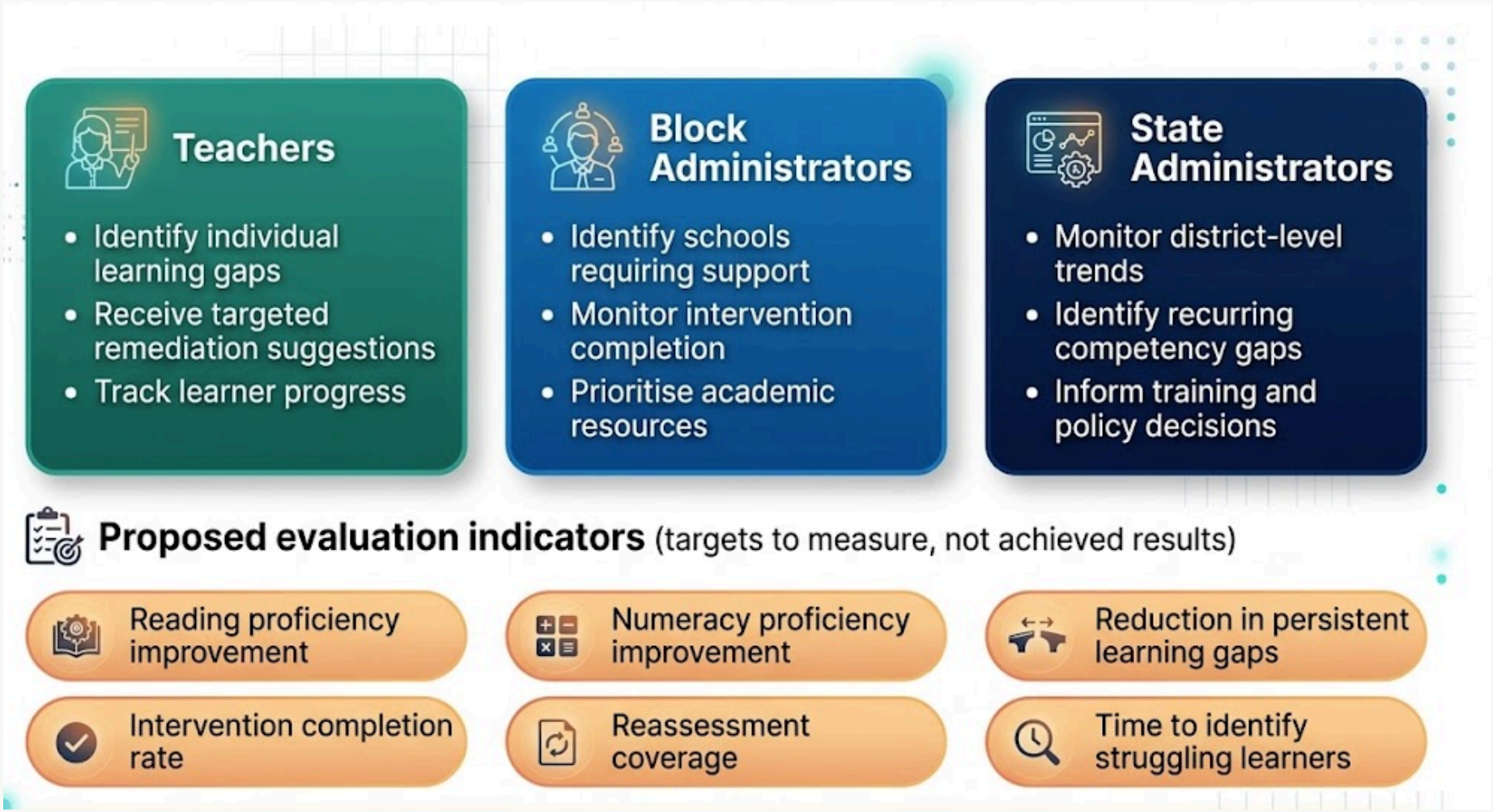
C. Data Governance

- Pseudonymous learner IDs
- Role-based access
- Data provenance
- Standardised schemas
- Audit logging
- Consent/authorisation and data minimisation

D. Feasibility

Component	Status
Dashboard	Prototype built
REST APIs	Implemented
FLN diagnostics	Implemented
Data ingestion	Prototype built
ML validation	Experimental
Government integration	Requires authorisation

From Data to Better Learning Outcomes



"NIPUNSetu does not seek to create another education data platform. It seeks to ensure that the data already being collected leads to timely, measurable and equitable learning interventions."